Chapter 1 exercises

These exercises are original. Unless stated otherwise, use a fixed-word RAM model. Hints identify tools without giving final results.

DS01-01 • The upper_bound contract and proof

For a nondecreasing array A, design binary search that returns the first position strictly greater than x. State the precondition, postcondition, invariant, and termination measure. Trace every iteration for `[2,4,4,9,13]`, x=4, and handle an empty array.

Hint. Define the inequalities satisfied by the excluded regions on each side.

DS01-02 • Nested-loop depth is not enough

For `n=2^k`, an outer loop takes i=n, n/2,…,1, with i inner iterations each time. Find the exact number of inner iterations and its tight bound. What changes if the inner loop always runs n times?

Hint. Write the sum before inferring anything from two nested loops.

DS01-03 • Growth by a noninteger factor

A dynamic array starts with capacity 1 and grows to `ceil(3c/2)` when full. Assume allocating and moving c elements costs O(c). Prove that n appends from empty cost O(n) in total. State the worst-case cost of one append.

Hint. Bound earlier capacities backward from the last expansion using geometric decay.

DS01-04 • Analysis types and arithmetic costs

An algorithm performs n arbitrary-precision integer additions. Every intermediate value has at most L bits. Give an upper bound for bitwise addition. If L grows with n, why is Θ(n) not automatically valid?

Hint. Separate iteration count from the cost of each iteration.

DS01-05 • An explicit potential for expensive high bits

A k-bit counter has low-bit index 0. Reads cost 1; writing bit i costs (3/2)^i; ignore other instructions. Start at zero and stop before overflow. Construct a nonnegative potential with no unspecified coefficient giving a uniform numerical amortized increment bound. Check the first increment that clears two bits. Use actual total cost to assess write costs 2^i when k varies.

Chapter 1 exercises (continued)

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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